

# Understanding Analysis, 2nd Edition

## Chapter 7 - Section 7.2 - The Riemann Integral

1. Let  $f$  be a bounded function on  $[a, b]$ , and let  $P$  be an arbitrary partition of  $[a, b]$ . First, explain why  $U(f) \geq L(f, P)$ . Now, prove Lemma 7.2.6: for any bounded function  $f$  on  $[a, b]$ , it is always the case that  $U(f) \geq L(f)$ .

**Solution:** We'll prove it by contradiction. Suppose that  $U(f) < L(f, P)$ . By definition of infimum, there must exist a partition  $P_2$  such that  $U(f, P_2) < L(f, P)$ , for otherwise the infimum would've been at least  $L(f, P)$ .

But  $U(f, P_2) < L(f, P)$  contradicts Lemma 7.2.4:  $L(f, P_1) \leq U(f, P_2)$  for any partitions  $P_1, P_2$ .

To use this fact to prove Lemma 7.2.6, note that  $L(f)$  is the supremum of all lower sums while  $U(f)$  is an upper bound. Since by definition, an upper bound is *at least* the supremum, it follows that  $U(f) \geq L(f)$ .

Alternatively, we can prove Lemma 7.2.6 directly by contradiction as well. Suppose  $U(f) < L(f)$  and let  $m$  be the midpoint i.e.  $m = \frac{L(f) + U(f)}{2}$ . By definition of infimum, there exists a partition  $P_1$  such that  $U(f, P_1) < m$ . By definition of supremum, there exists a partition  $P_2$  such that  $m < L(f, P_2)$ . But  $U(f, P_1) < m < L(f, P_2)$  contradicts Lemma 7.2.4.

2. Consider  $f(x) = 1/x$  over the interval  $[1, 4]$ . Let  $P$  be the partition consisting of the points  $\{1, 3/2, 2, 4\}$ .
  - (a) Compute  $L(f, P)$ ,  $U(f, P)$  and  $U(f, P) - L(f, P)$ .
  - (b) What happens to the value of  $U(f, P) - L(f, P)$  when we add the point 3 to the partition?
  - (c) Find a partition  $P'$  of  $[1, 4]$  for which  $U(f, P') - L(f, P') < 2/5$ .

**Solution:**

$$(a) \quad L(f, P) = (2/3 * 1/2) + (1/2 * 1/2) + (1/4 * 2) = 13/12.$$

$$U(f, P) = (1 * 1/2) + (2/3 * 1/2) + (1/2 * 2) = 11/6.$$

$$U(f, P) - L(f, P) = 3/4.$$

- (b) It gets smaller, because adding a new point to a partition decreases the upper sum while increasing the lower sum (Lemma 7.2.3). The new values are:

$$L(f, P) = (2/3 * 1/2) + (1/2 * 1/2) + (1/3 * 1) + (1/4 * 1) = 14/12.$$

$$U(f, P) = (1 * 1/2) + (2/3 * 1/2) + (1/2 * 1) + (1/3 * 1) = 5/3.$$

$$U(f, P) - L(f, P) = 1/2.$$

- (c) I initially solved this by adding more points to the initial partition + some random guessing, but chatgpt suggested a smarter way.

Let's create an equally-spaced partition. Then we have

$$U - L = \Delta x \sum_{i=1}^n M_i - m_i$$

For  $f(x) = 1/x$ , the supremum  $M_i = f(x_{i-1})$  and the infimum  $m_i = f(x_i)$ .

The sum telescopes:

$$\begin{aligned} U - L &= \Delta x [f(x_0) - f(x_1) + f(x_1) - f(x_2) + \dots + f(x_{n-1}) - f(x_n)] \\ &= \Delta x (f(x_0) - f(x_n)) \end{aligned}$$

We have  $x_0 = 1$  and  $x_n = 4$ , resulting in  $\Delta x(1 - 4) = \frac{3}{4}\Delta x$  and we want this to be less than  $\frac{2}{5}$ :

$$\frac{3}{4}\Delta x < \frac{2}{5} \implies \Delta x < \frac{8}{15}$$

We can pick  $\Delta x = \frac{1}{2}$ . So the final partition  $P'$  is  $\{1, \frac{3}{2}, 2, \frac{5}{2}, 3, \frac{7}{2}, 4\}$ .

### 3. (Sequential Criterion for Integrability)

- (a) Prove that a bounded function  $f$  is integrable on  $[a, b]$  if and only if there exists a sequence of partitions  $(P_n)_{n=1}^{\infty}$  satisfying

$$\lim_{n \rightarrow \infty} [U(f, P_n) - L(f, P_n)] = 0,$$

and in this case  $\int_a^b f = \lim_{n \rightarrow \infty} U(f, P_n) = \lim_{n \rightarrow \infty} L(f, P_n)$ .

- (b) For each  $n$ , let  $P_n$  be the partition of  $[0, 1]$  into  $n$  equal subintervals. Find formulas for  $U(f, P_n)$  and  $L(f, P_n)$  if  $f(x) = x$ . The formula  $1 + 2 + 3 + \dots + n = n(n+1)/2$  will be helpful.
- (c) Use the sequential criterion for integrability from (a) to show directly that  $f(x) = x$  is integrable on  $[0, 1]$  and compute  $\int_0^1 f$ .

#### Solution:

- (a) We'll prove the equivalence with the criterion of Theorem 7.2.8: for any  $\epsilon > 0$  there exists a partition  $P_\epsilon$  of  $[a, b]$  such that

$$U(f, P_\epsilon) - L(f, P_\epsilon) < \epsilon$$

For the sufficiency direction, assume the sequential criterion is true. Then by definition of sequential convergence, for any  $\epsilon > 0$ , there exists  $N$  such that for all  $n \geq N$

$$U(f, P_n) - L(f, P_n) < \epsilon$$

Then just set  $P_\epsilon = P_N$ .

For the necessity direction, assume Theorem 7.2.8 is true. Then we can build a sequence of partitions  $(P_n)$  where

$$U(f, P_n) - L(f, P_n) < 1/n$$

And since  $\lim_{n \rightarrow \infty} 1/n = 0$ , then  $\lim_{n \rightarrow \infty} U(f, P_n) - L(f, P_n) = 0$  as well.

— **This is incorrect** —

Finally, to prove that  $\int_a^b f = \lim_{n \rightarrow \infty} U(f, P_n) = \lim_{n \rightarrow \infty} L(f, P_n)$ , we need to prove that both limits are equal to  $U(f)$  and  $L(f)$  according to definition 7.2.1 of the Riemann Integral.

Note that by exercise 1 and the fact that  $U(f)$  and  $L(f)$  are infimum / supremum:

$$\lim_{n \rightarrow \infty} U(f, P_n) \geq U(f) \geq L(f) \geq \lim_{n \rightarrow \infty} L(f, P_n)$$

Since we know  $\lim_{n \rightarrow \infty} U(f, P_n) = \lim_{n \rightarrow \infty} L(f, P_n)$ , the only solution to the inequality is if they're equal  $U(f)$  and  $L(f)$ .

— **Mistake pointed out by chatgpt. Correction by my own.** —

The mistake is assuming that both  $\lim_{n \rightarrow \infty} U(f, P_n)$  and  $\lim_{n \rightarrow \infty} L(f, P_n)$  exist, which is not given and is part of what we need to prove.

We can prove it directly using the definition of sequential limit. We're given  $\lim_{n \rightarrow \infty} [U(f, P_n) - L(f, P_n)] = 0$ . Choose  $N$  large enough such that for every  $n \geq N$

$$U(f, P_n) - L(f, P_n) < \epsilon$$

The same  $N$  works to prove that  $\lim_{n \rightarrow \infty} U(f, P_n) = U(f)$  because  $U(f, P) \geq U(f) \geq L(f, P)$  for any  $P$ , hence

$$U(f, P_n) - U(f) \leq U(f, P_n) - L(f, P_n) < \epsilon$$

A similar argument can be used to show that  $\lim_{n \rightarrow \infty} L(f, P_n) = L(f)$ . Now we can apply the Algebraic Limit Theorem to conclude that  $\lim_{n \rightarrow \infty} [U(f, P_n) - L(f, P_n)] = 0 \implies \lim_{n \rightarrow \infty} U(f, P_n) = \lim_{n \rightarrow \infty} L(f, P_n)$ .

(b)

$$U(f, P_n) = \frac{1}{n} \sum_{i=1}^n \frac{i}{n} = \frac{1}{n^2} \sum_{i=1}^n i = \frac{n+1}{2n}$$

$$L(f, P_n) = \frac{1}{n} \sum_{i=1}^n \frac{i-1}{n} = \frac{1}{n^2} \sum_{i=1}^n (i-1) = \frac{n+1}{2n} - \frac{1}{n}$$

(c) We have  $\lim_{n \rightarrow \infty} U(f, P_n) - L(f, P_n) = \lim_{n \rightarrow \infty} \frac{1}{n} = 0$ , so the sequential criterion is satisfied.

To compute  $\int_0^1 f$ , we note that

$$U(f, P_n) = \frac{n+1}{2n} = \frac{1}{2} + \frac{1}{2n}$$

Hence the area is

$$\lim_{n \rightarrow \infty} U(f, P_n) = \frac{1}{2}$$

Similarly,  $\lim_{n \rightarrow \infty} L(f, P_n) = \frac{1}{2}$  as well, although we don't really need to check this since the sequential criterion guarantees that  $\lim_{n \rightarrow \infty} U(f, P_n) = \lim_{n \rightarrow \infty} L(f, P_n)$ .

Note that the result agrees with the geometric area as well; the area under  $f(x) = x$  from  $[0, 1]$  is a triangle with base and height 1, so its area is  $\frac{1}{2}$ .